

Course Code: 22CS C01

Course Title: PROBLEM SOLVING AND PROGRAMMING

Question Bank

UNIT I: Introduction to Programming - Evolution of languages: Machine, Assembly and High-level languages. Software requirements for programming: OS, compiler, linker, loader, editor. Design specification: Algorithms and Flowcharts.

Essay type questions

1. Define programming. What is a programming language? [CO1][BT1]
2. Write short note on generations of programming languages.[CO1][BT2]
3. How is third generation programming language is better than its predecessors.[CO1][BT4]
4. Explain Software requirements for programming.
5. Write a short note on the following.
OS, compiler, linker, loader, editor
6. Differentiate Compiler and Interpreter.
7. Define an Algorithm.
8. What is the importance of flowchart?
9. Examples on algorithm and flowchart
 - a. sum of first n natural numbers
 - b. addition of two numbers
 - c. largest number among three numbers.
 - d. Factorial of a number
 - e. Fibonacci series
 - f. Armstrong number

UNIT II: Data Types and Operators, Variable, Sequences and Iteration - Data types, Expressions, Precedence Rules, Operators: arithmetic, relational, logical, bitwise and miscellaneous operators; local variable, global variables, List, String, Tuples, Sequence mutation and accumulating patterns.

1. What is implicit conversion? give an example.[CO2][BT5]
2. Write a short note on data types in python.[CO1][BT2]
3. Explain type conversion with an example.[CO2][BT3]
4. Explain different operators (arithmetic, relational, logical, bitwise, membership and Identity) with an example.[CO2][BT3]
5. Write short notes on expressions and Precedence Rules in python.[CO2][BT2]
6. Explain operations on strings with an example.[CO2][BT3]
7. Explain Local and Global variables.
8. Write a brief notes on the following
 - a. List
 - b. String
 - c. Tuples
 - d. Sequence mutation and accumulating patterns.

UNIT III: Conditional Statement, Loops, Arrays and Strings, user-defined Data Types - if..else, for, while, nested iteration, Concept and use of arrays, declaration and usage of arrays, 2-dimensional arrays, different types of user defined data types.

Essay type questions

1. When should we use 'nested if' statements? Illustrate your answer with an example. [CO2][BT1]
2. Write a short note on conditional branching statements supported by python. [CO2][BT2]
3. Explain the syntax of for loop with an example. [CO1][BT2]
4. Is it necessary for every if block to be accompanied with an else block. Comment on this with an example.[CO2][BT4]
5. Explain the utility of Range() with the help of an example. [CO2][BT5]
6. For loop is usually known as a definite or determine loop. Justify the statement with the help of an example. [CO2][BT4]
7. Explain the utility of break statement with the help of an example. [CO1][BT2]
8. Differentiate between counter controlled loops and sequential controlled loops. [CO2][BT4]
9. Write a short note on nested loops. [CO2][BT2]
10. Explain the utility of continue statement with the help of an example. [CO1][BT2]
11. Differentiate between pass and continue. [CO1][BT4]
12. Write a short note on elif statement with an example. [CO1][BT2]
13. Write short notes on iterative statements. [CO1][BT2]
14. 'The return statement is optional'. Justify this statement with an example. [CO1][BT5]
15. Differentiate local and global variables. [CO1][BT1]
16. When you can have a variable with the same name as that of a global variable in the program, how is the name resolved in python? [CO2][BT3]
17. Explain the use of a return statement. [CO1][BT2]
18. What are the modules? How do you use them in your programs ? [CO2][BT3]
19. Explain about Arrays and declaration and use of arrays?

20. Explain about 2-dimensional arrays with an example.
21. Write short notes on Strings ?
22. Write short notes on user-defined Data Types
23. -if..else, for, while, nested iteration–Programs
24. Write short notes on different types of user defined data types.

UNIT IV: Dictionaries and Dictionary Accumulation, Functions/Methods - Dictionary basics, operations, methods, accumulation, advantages of modularizing program into functions, function definition and function invocation. Positional parameters passing arrays to functions, recursion, library functions.

1. Explain about Dictionaries and Dictionary Accumulation ?
2. Explain about operations, methods of Dictionaries ?
3. What are the advantages of modularizing programs into functions ?
4. Define function with an example and give its advantages. [CO2][BT1]
5. How to invoke functions. Explain with an example.
6. What are user defined functions? Give an example. [CO2][BT1]
7. What is modularization?
8. What do you understand by the term arguments? How do you pass them to a function.
[CO2][BT2]
9. Define a function that calculates the sum of all numbers from 0 to its argument.
[CO2][BT1]
10. Can a function call another function? justify your answer with an example. [CO1][BT5]
11. Write short notes on [CO2][BT2]
 - a) Keyword arguments
 - b) Default arguments
 - c) Positional arguments
 - d) Variable length arguments
11. How can we pass arrays to functions. Explain with an example.
12. Explain recursion with an example.
13. Explain about library functions with examples.

UNIT V: File Handling and Memory Management - Concepts of files and basic file operations, writing/reading data to/from a .csv file, Memory Management Operations.

1. Explain about File Handling and different types of file access modes ?
2. Explain about basic file operations ?
3. Explain about writing/reading data to/from a .csv file ?
4. Explain about Memory Management Operations ?